ASIAN UNIVERSITY FOR WOMEN

Department of Computer Science

CSCI 3000: Computer Communication Networks

Project On

Bank Enterprise Network: Design and Implementation Using CISCO Packet Tracer

Group 2

10th April, 2026

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Table of Content

1. Introduction
    1.1 Enterprise Network
    1.2 Objective
2. Network Topology
    2.1 Topology Diagram
    2.2 Network Components
3. IP Addressing Scheme
    3.1 Subnetting design
    3.2 IP address allocation
4. Device Configuration
5. Network Services
    5.1 DHCP
    5.2 DNS
    5.3 Web Server
    5.4 FTP Server
    5.5 NAT
6. Testing and Verification
7. Conclusion

Bank Enterprise Network

1. Introduction

A network is the interconnection of two or more computers, printers or nodes, other devices that are connected together to share the resources and allow electronic communication. These nodes can be connected to each other through a wired or wireless connection, and use protocols like IP protocol to transmit the data and information. There are different types of networks i.e Local Area Network(LAN), Wide Area Network(WAN), Metropolitan Area Network(Man), Personal Area Network(PAN) etc. These different types of Network are interconnected and integrated together to form an enterprise network.

An enterprise network is a larger scale, unified system of communication and IT resource that enables data sharing and access across an organization. It basically connects clients, employees, and stakeholders etc regardless of their location ensuring secure and reliable access to the company resources.The computers and devices on a network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams. This project focuses on the design and implementation of a bank enterprise network using Cisco Packet Tracer.

The modern banking system heavily depends on computer networks for real time communication, data transfer, processing transactions in real time, and data management. Therefore, the bank enterprise network integrates different types of network topologies like mesh and star, and follows a hierarchical network architecture, consisting of core, distribution and access layers. The network includes essential devices such as routers, switches, servers, access points, printers, PC's,etc along with security mechanisms like Vlans and access control list to ensure confidentiality and integrity in banking data.

Objective

The objective of this bank enterprise networks are:

- To provide easier network management to the administrator through centralized management
- Reliable communication across the network in the bank
- Easier addition and deletion of nodes, scalability
- To provide a secure network for data transmission
- High availability and fault tolerance
- Efficient resources sharing
- Mobility and flexibility
- Secure data transmission
- Segmentation for better performance

2. Network Topology

A network topology is the arrangement of devices (nodes) and connections (links) in a computer network. It shows how computers, servers, and other devices are connected and how data flows between them.

There are different types of topologies like mesh topology, star topology, bus topology, ring topology, tree topology, and hybrid topology. The bank enterprise works and is designed under the standard of the hybrid topology. It consists of start, tree and mesh topologies to ensure scalability, reliability, security and high availability.

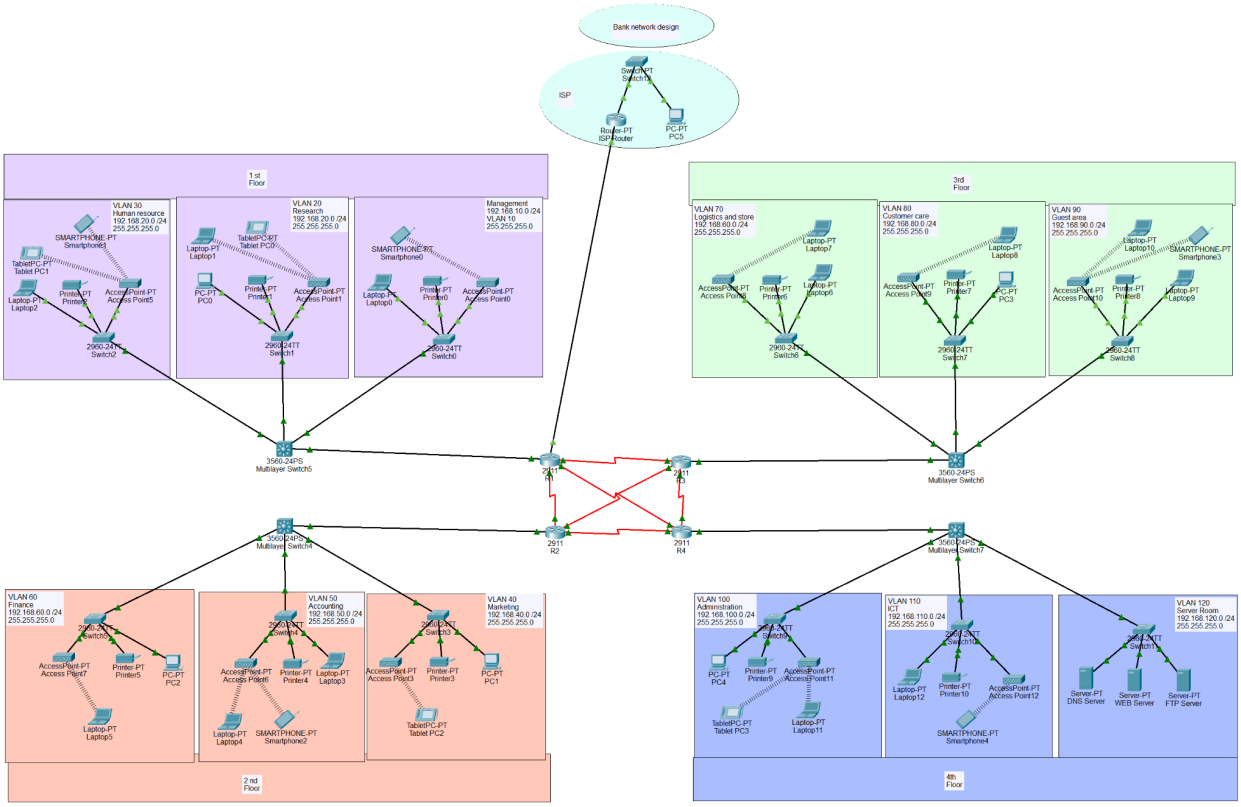

Figure of: Bank Enterprise Network

2.1 Network Components

Switch:

Switch is a hardware network device that connects multiple devices like computers, printers and servers within a network. In the bank enterprise network, switches are deployed to connect each department such as finance, accounting, marketing, human resources, etc, to enable efficient and simultaneous local communication. Here Layer 3 multilayer switch, 2960 TT switch is used. They also support VLANs for improved security, traffic management and segmentation. It operates at Layer 2 (Datalink layer) of the OSI model and uses MAC addresses to forward data frames to designated destinations. When a device sends the data, the switch learns and records MAC address and incoming port number in its MAC address table. By using the table and observing the source MAC address of incoming frames the switch efficiently forwards them to its intended port, reducing the unnecessary traffic and enhancing the network performance.

Routers:

A router is the device that helps to connect multiple networks or sub networks together and allows them to communicate with each other. Its primary function is to manage the traffic between these networks by forwarding the data packets to their designated destination IP address and allowing multiple devices to use the same internet connection. To direct packets effectively, a router uses an internal routing table, a list of paths to various network destinations. When a packet arrives, the router reads a packet's header to determine its destination, then consults the routing table to figure out the most efficient path to that destination. It then forwards the packet to the next network along the path. The bank enterprise operates on Wide Area Network (WAN) which consists of multiple routers connecting different departments. These routers not only form the network but also manage the flow of data within and between the branches, ensuring that information is transmitted securely and efficiently across the enterprise network.

Servers:

A server refers to a powerful computer designed to function as the central hub to manage network resources and provide services to other computers, known as clients, in a network setting. It handles network tasks such as data storage, application hosting, email processing and more. This bank enterprise network uses generic multi-service servers(Server-PT) placed in a dedicated Server Room VLAN, which provides essential services such as database management, DHCP, DNS, application hosting and data backup ensuring secure and centralized banking operation. Servers operates at the application layer(Layer 7) of the OSI model while relying on lower layers of the data transmission across the network.

PC’s/ Laptops (Clients devices):

PCs, laptops and tablets are the end-user devices, functioning as a network node, connected through a wired or wireless media designed to access and interact with network resources and utilize the services provided by servers. These are the primary interface for bank staff to perform essential operations, such as transaction processing, email communication, database queries and running applications, etc. This hardware operates in the Layer 7, application layer of the OSI model and relies on lower layers for communication and data transmission. In this network, these devices are connected through dedicated VLANs for each department, ensuring secure access to resources while maintaining proper network segmentation.

Access Point:

Access point is a wireless network device that acts as a bridge between wireless client devices and wired local area network (LAN). It receives data from the network through an ethernet connection and transmits

it as Wi-Fi signals for wireless devices, and vice versa. In enterprise networks, access points are typically powered using PoE (Power over Ethernet). Which allows both data and electric power to be delivered through a single cable from router or switch. Access points extend the wireless coverage of the existing network, allowing more concurrent users to connect and seamless access to network resources throughout the organization. Access points primarily operate at the Layer 2, data link layer of the OSI model and handles MAC addressing, frame forwarding and wireless accessing. In the bank enterprise network, access points extend wireless coverage, support multiple simultaneous users and provide secure, seamless access to applications, databases,and other network resources throughout the organization.

Network Components	Count	Function
Switch	12-16	Department Connectivity & VLAN support
Multi layer switch	4	Core aggregation & routing
Router	5	Network interconnection
Servers	3	Client services & data management
Access Points	12	Wireless coverage & mobility
End Devices	24-30 (extendable up to 3038 users)	User devices

3. IP Addressing Scheme

3.1 Subnetting Design

The bank enterprise network uses a structured and hierarchical IP addressing scheme to ensure efficient communication, flexibility and security. It has a total of 12 departments with 3 in each floor. The network is divided into multiple VLANs, where each VLAN represents a specific department such as VLAN 10 indicates Management department.

The subnetting design is divided into three distinct categories based on the function of each link:

- VLAN/Department Networks → 192.168.X.0/24
- Router-to-Router WAN Links → 10.10.10.X/30
- Router-to-Floor Switch Links → 10.0.X.0/30
- ISP/Internet Connection → 200.0.0.0/30

For departmental networks, a /24 subnet mask (255.255.255.0) is used. The subnet uses the Class C private IP range 192.168.X.0/24, where X represents the corresponding VLAN ID. This provides 256 total IP addresses with 254 usable host IP addresses per VLAN, which is sufficient for devices such as PCs, laptops, printers, and mobile devices within each department.

For point-to-point communication between routers, /30 subnet mask (255.255.255.252) is used. It provides exactly 4 total IPs with 2 usable host IP addresses, making them ideal for router-to-router links and minimizing IP address wastage. The 10.10.10.0/30 block is subdivided into /30 chunks for each router-to-router serial link and the private IP range 10.0.X.0/30 block for router-to-floor-switch Gigabit link .

Additionally, the connection between router R1 and the ISP also uses a /30 subnet on the public IP range 200.0.0.0/30. This is standard practice for ISP uplinks since only 2 addresses are needed (one for R1 and one for the ISP router). It enables external communication as well as wastes the fewest public IP addresses possible.

This subnetting approach ensures:

1. Better performance and security
2. Easy scalability for future expansion.
3. Logical network segmentation
4. Efficient IP usage.

VLAN Subnet Allocation :

VLAN	Department	Network Address	Subnet Mask	Usable Hosts
10	Management	192.168.10.0/24	255.255.255.0	254
20	Research	192.168.20.0/24	255.255.255.0	254
30	HR	192.168.30.0/24	255.255.255.0	254
40	Marketing	192.168.40.0/24/24	255.255.255.0	254
50	Accounting	192.168.50.0/24	255.255.255.0	254
60	Finance	192.168.60.0/24	255.255.255.0	254
70	Logistics	192.168.70.0/24	255.255.255.0	254
80	Customer Care	192.168.80.0/24	255.255.255.0	254
90	Guest	192.168.90.0/24	255.255.255.0	254

100	Administration	192.168.100.0/24	255.255.255.0	254
110	ICT	192.168.110.0/24	255.255.255.0	254
120	Server Room	192.168.120.0/24	255.255.255.0	254

WAN Subnet Allocation :

Link	Network	Subnet Mask	Usable Hosts
R1 – R2	10.10.10.0/30	255.255.255.252	2
R1 – R3	10.10.10.4/30	255.255.255.252	2
R1 – R4	10.10.10.8/30	255.255.255.252	2
R1- ISP (public)	200.0.0.0/30	255.255.255.252	2

3.2 IP Address Allocation

Table 1 - Core Router WAN Interface Addresses

Each router has one GigabitEthernet interface connecting it to its floor's Layer 3 multilayer switch using a /30 point-to-point subnet.

Device	Interface	IP Address	Subnet Mask	Description
R1	G0/0	10.0.1.1	255.255.255.252	To Floor 1-L3
R2	G0/0	10.0.2.1	255.255.255.252	To Floor 2-L3
R3	G0/0	10.0.3.1	255.255.255.252	To Floor 3-L3
R4	G0/0	10.0.4.1	255.255.255.252	To Floor 4-L3

Table 2 - Router-to-Router Serial Links

The four core routers are interconnected using serial links arranged in a partial mesh topology. R1 acts as the hub router, connecting directly to R2, R3, and R4. Each serial link uses a /30 subnet from the 10.10.10.0 block.

Link	Network	R1 IP	Other Router IP
R1 ↔ R2	10.10.10.0/30	10.10.10.1	10.10.10.2
R1 ↔ R3	10.10.10.4/30	10.10.10.5	10.10.10.6
R1 ↔ R4	10.10.10.8/30	10.10.10.9	10.10.10.10

Table 3 - ISP Connection Addresses

Router R1 is the internet gateway. It connects to the ISP router via the public subnet 200.0.0.0/30. An external PC at 8.8.8.2 simulates an internet user for connectivity testing.

Device	Interface	IP Address	Subnet Mask	Description
R1	G0/1	200.0.0.2	255.255.255.252	Link To ISP
ISP	Fa0/0	200.0.0.1	255.255.255.252	Link To R1
ISP	Fa1/0	8.8.8.1	255.255.255.0	External Network
External PC	Fa0	8.8.8.2	255.255.255.0	Internet User

Table 4 - VLAN Department Networks

Each of the 12 VLANs is assigned a /24 subnet from the 192.168.0.0 private address space. The gateway address (X.X.X.1) is configured as a Switched Virtual Interface (SVI) on the floor's multilayer switch and serves as the default gateway for all the devices in that VLAN.

VLAN	Department	Network Address	Subnet Mask	Gateway
10	Management	192.168.10.0/24	255.255.255.0	192.168.10.1
20	Research	192.168.20.0/24	255.255.255.0	192.168.20.1
30	HR	192.168.30.0/24	255.255.255.0	192.168.30.1
40	Marketing	192.168.40.0/24/24	255.255.255.0	192.168.40.1

50	Accounting	192.168.50.0/24	255.255.255.0	192.168.50.1
60	Finance	192.168.60.0/24	255.255.255.0	192.168.60.1
70	Logistics	192.168.70.0/24	255.255.255.0	192.168.70.1
80	Customer Care	192.168.80.0/24	255.255.255.0	192.168.80.1
90	Guest	192.168.90.0/24	255.255.255.0	192.168.90.1
100	Administration	192.168.100.0/24	255.255.255.0	192.168.100.1
110	ICT	192.168.110.0/24	255.255.255.0	192.168.110.1
120	Server Room	192.168.120.0/24	255.255.255.0	192.168.120.1

Table 5 - Server IP Addresses (VLAN 120 - Server Room)

Servers are assigned static IP addresses within VLAN 120. Static addressing ensures they are always reachable at the same address, which is required for DNS, web and FTP services.

Server	IP Address	Purpose
DNS Server	192.168.120.10	Domain resolution (bank.com to 192.168.120.20)
Web Server	192.168.120.20	Hosts the bank website (http://bank.com)
FTP Server	192.168.120.30	File transfer between departments

Table 6 - End Device IP Addresses :

All end-user devices automatically receive IP addresses via DHCP. Servers use static assignment. The DHCP pools are configured on each floor's multilayer switch.

Device	VLAN	IP Address	Gateway	Method
Laptop/Smartphone (Management)	VLAN 10	192.168.10.x	192.168.10.1	DHCP
Laptop/Tablet (Research)	VLAN 20	192.168.20.x	192.168.20.1	DHCP
Laptop/Smartphone (HR)	VLAN 30	192.168.30.x	192.168.30.1	DHCP

PC/Tablet (Marketing)	VLAN 40	192.168.40.x	192.168.40.1	DHCP
Laptop/Smartphone (Accounting)	VLAN 50	192.168.50.x	192.168.50.1	DHCP
PC/Laptop (Finance)	VLAN 60	192.168.60.x	192.168.60.1	DHCP
Laptops (Logistics)	VLAN 70	192.168.70.x	192.168.70.1	DHCP
PC/Laptop (Customer Care)	VLAN 80	192.168.80.x	192.168.80.1	DHCP
Guest Devices (Wireless)	VLAN 90	192.168.90.x	192.168.90.1	DHCP
Laptop (Admin)	VLAN 100	192.168.100.x	192.168.100.1	DHCP
PC (ICT)	VLAN 110	192.168.110.x	192.168.110.1	DHCP
Servers (Server Room)	VLAN 120	192.168.120.10-192.168.120.30	192.168.120.1	Static

4. Device Configuration

This section explains the key configurations applied to routers, switches, VLANs and the routing protocol used in the banking enterprise network. The network uses CISCO 2911 routers, CISCO 3560 multilayer (Layer 3) switches and Cisco 2960 Layer 2 access switches, all configured using Cisco IOS command-line interface (CLI).

1. Router configuration:

Four Cisco 2911 routers (R1, R2, R3, R4) form the core routing backbone of the network. Each router is connected to its floor's multilayer switch through GigaEthernet0/0 and to the other routers via serial links.

Router functions:

- Assign IP addresses to interfaces
- Enable communication between VLANs (Inter-VLAN Routing)
- Connect different floors and networks
- Establish WAN links between routers

Key configurations includes:

- Interface IP addressing (ip address)
- Activating interfaces (no shutdown)

- Serial link configuration
- Default static route to ISP
- OSPF routing configuration

R1 Configuration (Gateway Router — connects to ISP)

```
Router> enable
Router# configure terminal
Router(config)# hostname R1

! Interface to Floor 1 Multilayer Switch
R1(config)# interface GigabitEthernet0/0
R1(config-if)# ip address 10.0.1.1 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# exit

! Interface to ISP
R1(config)# interface GigabitEthernet0/1
R1(config-if)# ip address 200.0.0.2 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# exit

! Serial link to R2
R1(config)# interface Serial0/0/0
R1(config-if)# ip address 10.10.10.1 255.255.255.252
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
R1(config-if)# exit

! Default route to ISP (for internet access)
R1(config)# ip route 0.0.0.0 0.0.0.0 200.0.0.1
```

Description: The *'ip address'* command assigns an IP address to the interface. The *'no shutdown'* command activates the interface. The GigabitEthernet0/0 interface connects R1 to the multilayer switch using a /30 subnet for point-to-point communication. The GigabitEthernet0/1 connects R1 to the ISP using a public IP range. The serial interface connects it to other routers to form WAN links. The default route (0.0.0.0/0) forwards all unknown traffic to the ISP through internet access to reach external networks.

R2/R3/R4 CONFIGURATION

R2, R3, and R4 follow the same pattern. Only the interface addresses differ based on the allocation tables above.

```
Router> enable
Router# configure terminal
Router(config)# hostname R2
```

```
! Interface to Floor 2 Multilayer Switch
R2(config)# interface GigabitEthernet0/0
R2(config-if)# ip address 10.0.2.1 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# exit

! Serial link to R1
R2(config)# interface Serial0/0/0
R2(config-if)# ip address 10.10.10.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# exit
```

Description: Each router connects to its floor's multilayer switch using a GigabitEthernet interface. This allows high-speed communication between router and local network of that floor. The serial interface interconnects the routers using /30 subnetting which enables communication between different floors of the network.

2. Switch Configuration

Two types of switches are used in this network. Access switches in Layer 2 to connect end devices to their VLAN for each department. Multilayer switches per floor in Layer 3, to carry trunk links between floors and perform inter-VLAN routing.

Switch functions :

- Assign ports to specific VLANs
- Enable communication within VLANs
- Connect to routers using trunk links

Key configurations include:

- Access ports for end devices
- Trunk ports for inter-switch/router communication
- VLAN assignment

LAYER 3 MULTILAYER SWITCH --- FLOOR 1 EXAMPLE

The multilayer switch for each floor performs inter-VLAN routing. It must have IP routing enabled, VLAN interfaces (SVIs) configured for each VLAN, and a trunk uplink to the floor router.

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname MLS-Floor1

! Enable Layer 3 routing
MLS-Floor1(config)# ip routing
```

```
! Trunk port to Router R1
MLS-Floor1(config)# interface GigabitEthernet0/1
MLS-Floor1(config-if)# no switchport
MLS-Floor1(config-if)# ip address 10.0.1.2 255.255.255.252
MLS-Floor1(config-if)# no shutdown
MLS-Floor1(config-if)# exit

! Trunk ports to access switches
MLS-Floor1(config)# interface GigabitEthernet0/2
MLS-Floor1(config-if)# switchport trunk encapsulation dot1q
MLS-Floor1(config-if)# switchport mode trunk
MLS-Floor1(config-if)# no shutdown
MLS-Floor1(config-if)# exit
```

Description : The *'ip routing'* command enables Layer 3 functionality on the multilayer switch allowing routing between different VLANs. The *'no switchport'* command converts the interface into a routed port. Instead of acting like a normal switch port, it behaves like a router interface and is used to directly connect the multilayer switch to the router.This connection between the switch and router is a Layer 3 link, not a trunk, because it carries only one network and does not transport multiple VLANs.

On the other hand, trunk ports are used to connect switches together or connect access switches to the multilayer switch. These ports use IEEE 802.1Q encapsulation to carry traffic from multiple VLANs over a single physical link.

This overall configuration ensures efficient communication between switches, proper routing between VLANs, and a well-structured network design.

LAYER 2 ACCESS SWITCH --- HR department Example

Each department switch is set up to place its ports into the correct VLAN and connect to the floor's multilayer switch via a trunk port.

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW-HR

! Assign access ports to VLAN 30 (HR department)
SW-HR(config)# interface range FastEthernet0/1 - 10
SW-HR(config-if-range)# switchport mode access
SW-HR(config-if-range)# switchport access vlan 30
SW-HR(config-if-range)# exit
Trunk configuration (uplink to multilayer switch)
! Uplink to Multilayer Switch — must be trunk
SW-HR(config)# interface GigabitEthernet0/1
SW-HR(config-if)# switchport mode trunk
```

```
SW-HR(config-if)# no shutdown
SW-HR(config-if)# exit
```

Description : Access ports connect end devices and are assigned to a single VLAN. In this example, VLAN 30 represents the HR department so that all HR devices are logically grouped and secured within their department. The trunk port connects the access switch to the multilayer switch, allowing multiple VLANs to pass through. This allows different departments to communicate through the network.

3. VLAN Configuration

VLANs are implemented to logically separate departments within the same physical network. Devices in different VLANs cannot communicate directly unless routing is enabled.

Features of VLAN :

- Represents a specific department
- Has its own IP subnet
- Improves security and reduces broadcast traffic

Example:
VLAN 10 → Management
VLAN 30 → HR
VLAN 60 → Finance

The following configuration is applied on each multilayer switch to create VLANs and their gateway interfaces (SVIs):

```
MLS-Floor1(config)# vlan 10
MLS-Floor1(config-vlan)# name Management
MLS-Floor1(config-vlan)# exit

MLS-Floor1(config)# vlan 20
MLS-Floor1(config-vlan)# name Research
MLS-Floor1(config-vlan)# exit

MLS-Floor1(config)# vlan 30
MLS-Floor1(config-vlan)# name HumanResource
MLS-Floor1(config-vlan)# exit
```

Description : VLANs divide a physical network into separate broadcast domains. This helps to improve network performance and security. Devices in different VLANs cannot communicate directly without routing.

SVI (Gateway Interfaces)

```
! Create SVI (gateway) for each VLAN
MLS-Floor1(config)# interface vlan 10
MLS-Floor1(config-if)# ip address 192.168.10.1 255.255.255.0
MLS-Floor1(config-if)# no shutdown
MLS-Floor1(config-if)# exit

MLS-Floor1(config)# interface vlan 20
MLS-Floor1(config-if)# ip address 192.168.20.1 255.255.255.0
MLS-Floor1(config-if)# no shutdown
MLS-Floor1(config-if)# exit

MLS-Floor1(config)# interface vlan 30
MLS-Floor1(config-if)# ip address 192.168.30.1 255.255.255.0
MLS-Floor1(config-if)# no shutdown
MLS-Floor1(config-if)# exit
```

Description : SVIs act as the default gateway for each VLAN which allows devices to communicate outside their own network. The .1 address is used as the gateway for devices in each subnet. They enable inter-VLAN routing on the multilayer switch. Without SVIs, communication between VLANs would not be possible.

4. Routing Protocol - OSPF

The network uses OSPF (Open Shortest Path First) as the dynamic routing protocol. OSPF is a link-state routing protocol where each router builds a complete map of the network and uses the Shortest Path First (SPF) algorithm to find the best route to every destination.

Why is OSPF used?

- Open standard as it works across different vendors, not Cisco-only
- Efficient routing between multiple routers
- Fast convergence
- Automatic route updates
- Selects best path based on cost (bandwidth)
- Better performance compared to static routing
- Scalable for large enterprise networks like bank

OSPF Configuration on Each Router (example: R1)

```
R1(config)# router ospf 1

! Advertise all internal VLAN networks
R1(config-router)# network 192.168.0.0 0.0.255.255 area 0
```

```
! Advertise WAN router-to-router links
R1(config-router)# network 10.0.0.0 0.255.255.255 area 0

! Advertise the default route to other routers (R1 only)
R1(config-router)# default-information originate
R1(config-router)# exit
```

Description : The '*router ospf 1*' starts the OSPF routing process. The '*network*' command advertises connected networks so that other routers can learn them. Wildcard masks are the inverse of subnet masks. For example : 0.0.255.255 includes all in the range 192.168.X.X. OSPF uses the Shortest Path First (SPF) algorithm to determine optimal routes. The *'default-information originate'* command allows R1 to advertise the default route to other routers for internet access.

OSPF Configuration on Multilayer Switches

```
MLS-Floor1(config)# router ospf 1
MLS-Floor1(config-router)# network 192.168.10.0 0.0.0.255 area 0
MLS-Floor1(config-router)# network 192.168.20.0 0.0.0.255 area 0
MLS-Floor1(config-router)# network 192.168.30.0 0.0.0.255 area 0
MLS-Floor1(config-router)# network 10.0.1.0 0.0.0.3 area 0
MLS-Floor1(config-router)# exit
```

Description : The multilayer switch uses OSPF to advertise its VLAN networks to other routers. This allows routing between different floors and departments. It ensures all routers automatically receive updated routing information to make the network dynamic.

5. Network Services

The bank enterprise network implements several critical services to support daily banking operations across all departments. These services are hosted on three dedicated servers located in the Server Room (VLAN 120) with static IP addresses, ensuring that they remain consistently reachable from any part of the network.

5.1 DHCP (Dynamic Host Configuration Protocol)

DHCP is used to automatically assign IP addresses and network configuration parameters to all end-user devices across the enterprise. Rather than manually configuring each device, the DHCP service ensures that every PC, laptop, smartphone, and tablet receives a correct IP address, subnet mask, default gateway, and DNS server address when it joins the network.

In this network, DHCP pools are configured on each floor's multilayer switch (Layer 3 switch). Each pool corresponds to a specific VLAN and department, so devices always receive addresses within the correct subnet for their department.

DHCP Pool Configuration per VLAN:

VLAN	Department	Network	DHCP Range	Subnet Mask	Default GW
10	Management	192.168.10.0/24	10.2 – 10.254	255.255.255.0	192.168.10.1
20	Research	192.168.20.0/24	20.2 – 20.254	255.255.255.0	192.168.20.1
30	HR	192.168.30.0/24	30.2 – 30.254	255.255.255.0	192.168.30.1
40	Marketing	192.168.40.0/24	40.2 – 40.254	255.255.255.0	192.168.40.1
50	Accounting	192.168.50.0/24	50.2 – 50.254	255.255.255.0	192.168.50.1
60	Finance	192.168.60.0/24	60.2 – 60.254	255.255.255.0	192.168.60.1
70	Logistics	192.168.70.0/24	70.2 – 70.254	255.255.255.0	192.168.70.1
80	Customer Care	192.168.80.0/24	80.2 – 80.254	255.255.255.0	192.168.80.1
90	Guest	192.168.90.0/24	90.2 – 90.254	255.255.255.0	192.168.90.1
100	Administration	192.168.100.0/24	100.2 –100.254	255.255.255.0	192.168.100.1
110	ICT	192.168.110.0/24	110.2 –110.254	255.255.255.0	192.168.110.1
120	Server Room	192.168.120.0/24	Static only	255.255.255.0	192.168.120.1

The screenshot below shows **DHCP** successfully assigning IP addresses to devices in the network:

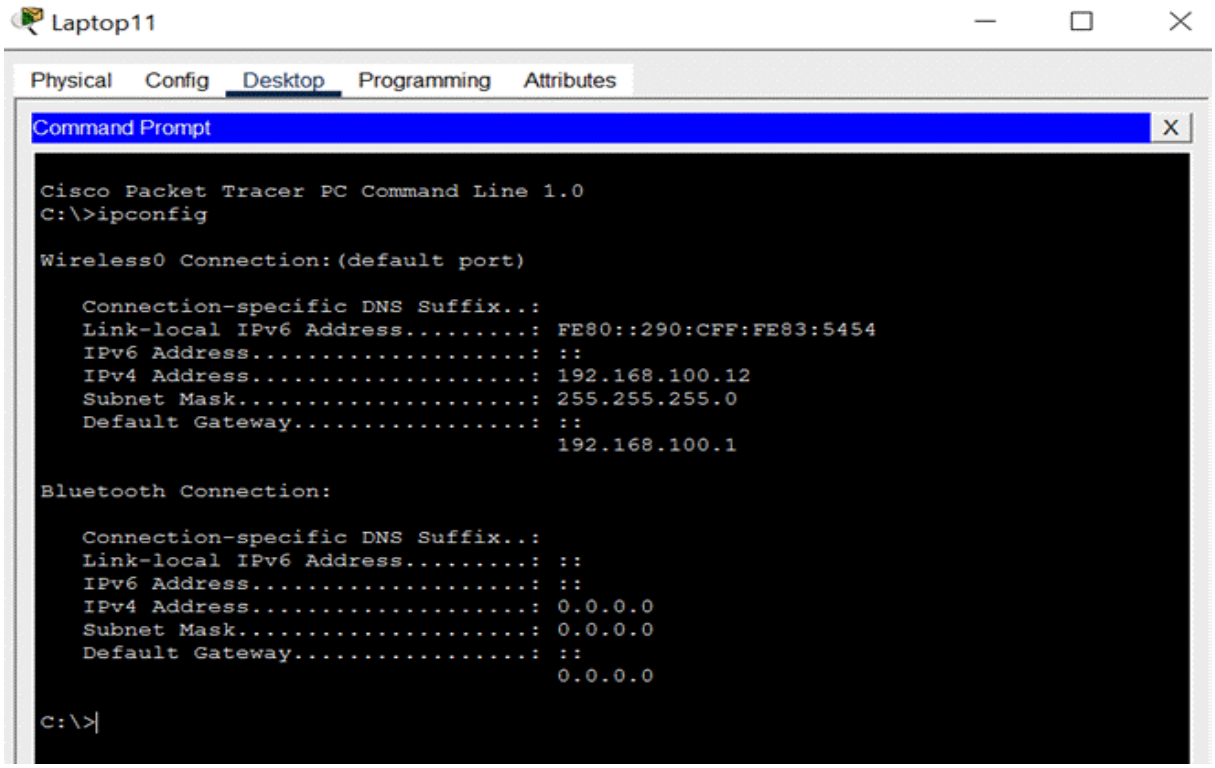

Figure 5.1 – DHCP address assignment verified in Cisco Packet Tracer

5.2 DNS (Domain Name System)

A DNS server is hosted at 192.168.120.10 in the Server Room (VLAN 120). It resolves the internal domain name bank.com to the web server's IP address (192.168.120.20), allowing users across all departments to access the bank's intranet portal using a readable name rather than a numeric IP address.

DNS service details:

- DNS Server IP: 192.168.120.10 (static, VLAN 120)
- Domain name: bank.com → resolves to 192.168.120.20 (Web Server)
- All DHCP pools include DNS server IP so devices automatically use this server for name resolution
- Provides forward lookup: hostname to IP address

The screenshot below confirms that the **DNS** server successfully resolves bank.com:

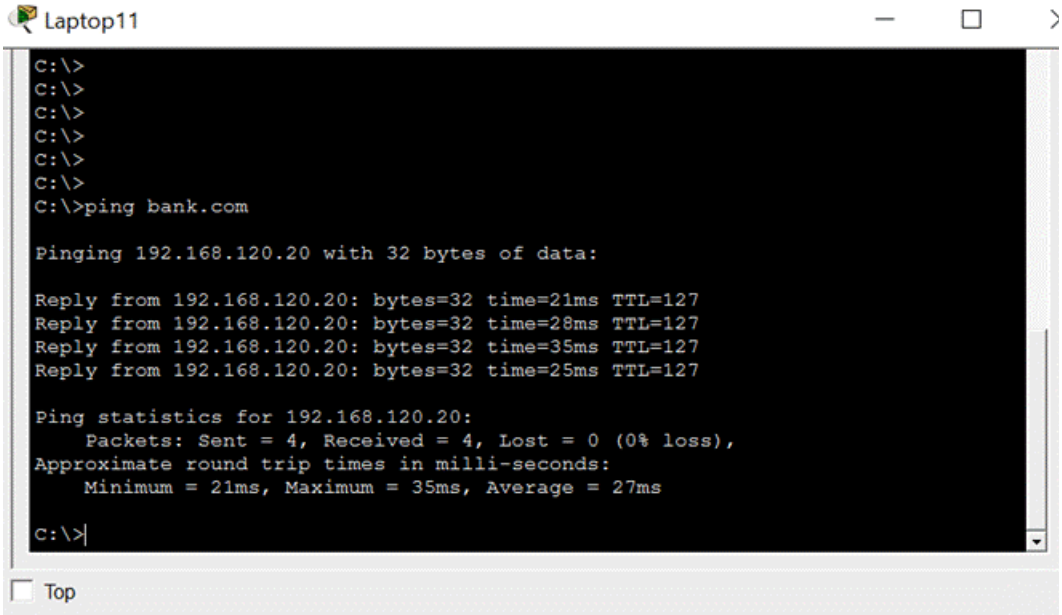

Figure 5.2 – DNS resolution of bank.com verified

5.3 Web Server

A web server is hosted at 192.168.120.20 (VLAN 120) and serves the bank's internal website accessible at http://bank.com. The server provides the banking portal for staff to access internal dashboards, transaction reports, and department-specific web applications.

Web server details:

- Web Server IP: 192.168.120.20 (static, VLAN 120)
- Protocol: HTTP (port 80) configured in Cisco Packet Tracer
- Domain: bank.com (resolved via DNS server at 192.168.120.10)
- Accessible from all internal VLANs through inter-VLAN routing

The screenshot below shows successful web browser access to http://bank.com from a client in another VLAN:

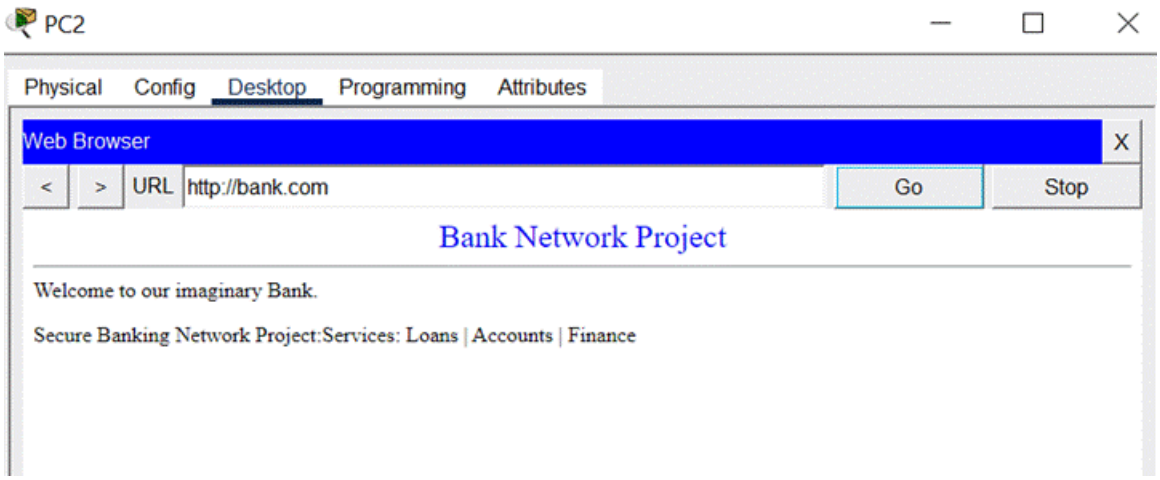

Figure 5.3 – Web server access to bank.com from a client device

5.4 FTP Server (File Transfer Protocol)

An FTP server is hosted at 192.168.120.30 (VLAN 120) to enable secure file transfer between departments within the enterprise. Bank staff use FTP to upload or download documents, reports, and data files to and from the central server.

FTP server details:

- FTP Server IP: 192.168.120.30 (static, VLAN 120)
- Protocol: FTP (port 21) configured in Cisco Packet Tracer
- Used by departments such as Finance, HR, and Accounting for document exchange
- Credentials (username/password) are set up on the server to restrict unauthorized access

The screenshot below demonstrates a successful **FTP** connection and file transfer from a client:

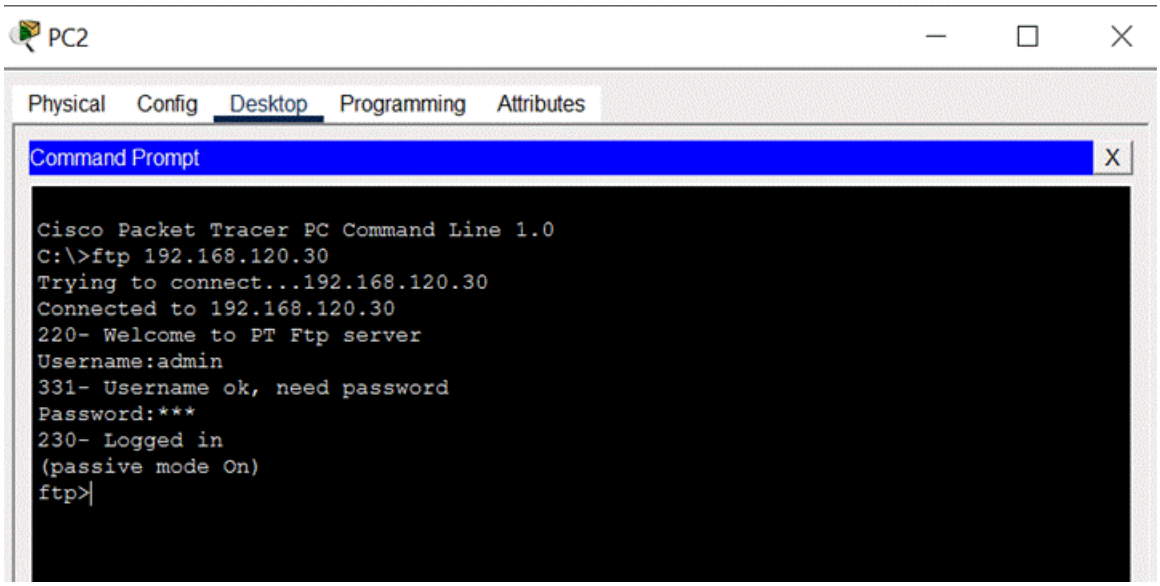

Figure 5.4 – FTP file transfer verified between client and FTP server

5.5 NAT (Network Address Translation)

Network Address Translation (NAT) is configured on Router R1, which acts as the internet gateway. NAT translates the private IP addresses of all internal devices (in the 192.168.0.0/16 range) into the single public IP address assigned to R1's interface facing the ISP (200.0.0.2). This allows all internal devices to access the internet without requiring individual public IP addresses.

The type of NAT used is PAT (Port Address Translation), also known as NAT Overload, where multiple internal hosts share one public IP address and are differentiated by unique port numbers.

NAT configuration on R1:

- Inside interface: GigabitEthernet0/0 (towards Floor 1 multilayer switch, private network)
- Outside interface: GigabitEthernet0/1 (towards ISP, public IP 200.0.0.2)
- Access Control List (ACL): permits all traffic from 192.168.0.0/16
- Command: ip nat inside source list <ACL> interface G0/1 overload
- Default route to ISP: ip route 0.0.0.0 0.0.0.0 200.0.0.1

The router configuration screenshot below shows the NAT, interface, and OSPF configuration on R1:

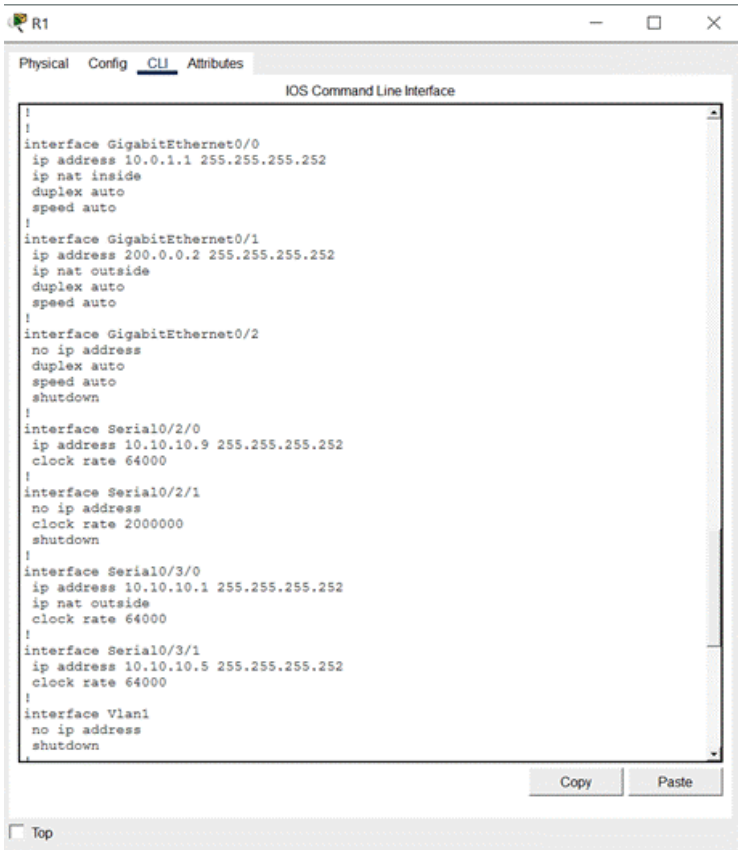

Figure 5.5a – Router R1 interface and NAT configuration

Figure 5.5b – Router configuration showing interfaces, OSPF, and NAT

6. Testing and Verification

This section documents the tests performed to verify that the bank enterprise network is functioning correctly. All tests were conducted using Cisco Packet Tracer in both real-time and simulation modes. The tests confirm DHCP assignment, DNS resolution, web and FTP server access, inter-VLAN connectivity, and internet access through NAT.

6.1 DHCP Verification

DHCP was tested by setting end devices in each VLAN to obtain their IP configuration automatically. The assigned addresses were verified to fall within the correct VLAN subnet, with the correct default gateway and DNS server information.

Results: All end devices successfully received DHCP-assigned IP addresses from their respective VLAN pools. The DHCP server on each multilayer switch correctly served requests from all departments.

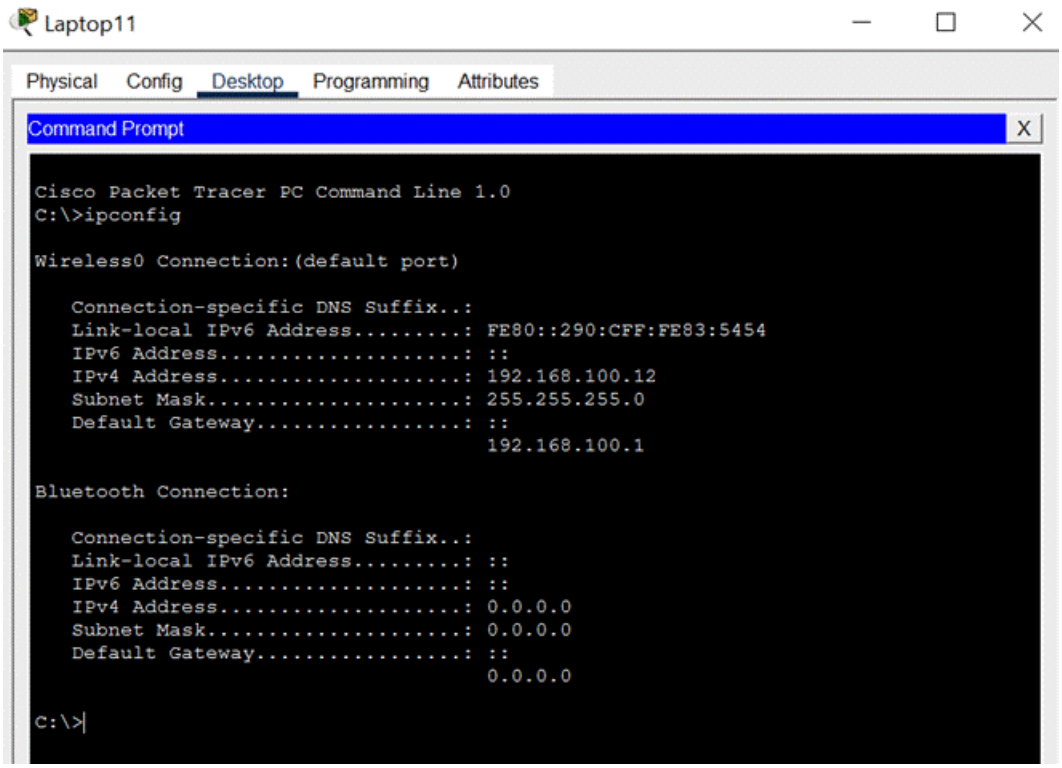

Figure 6.1 – DHCP assignment verified; device in VLAN 20 receives 192.168.20.x

6.2 DNS Resolution Test

DNS resolution was tested by opening a web browser on a client PC in a different VLAN and entering the URL http://bank.com. The DNS server at 192.168.120.10 resolved this domain to the web server's IP (192.168.120.20), and the web page loaded successfully.

Results: The DNS server correctly resolved bank.com across all tested VLANs, confirming that DHCP-distributed DNS server settings are working and DNS infrastructure is functional.

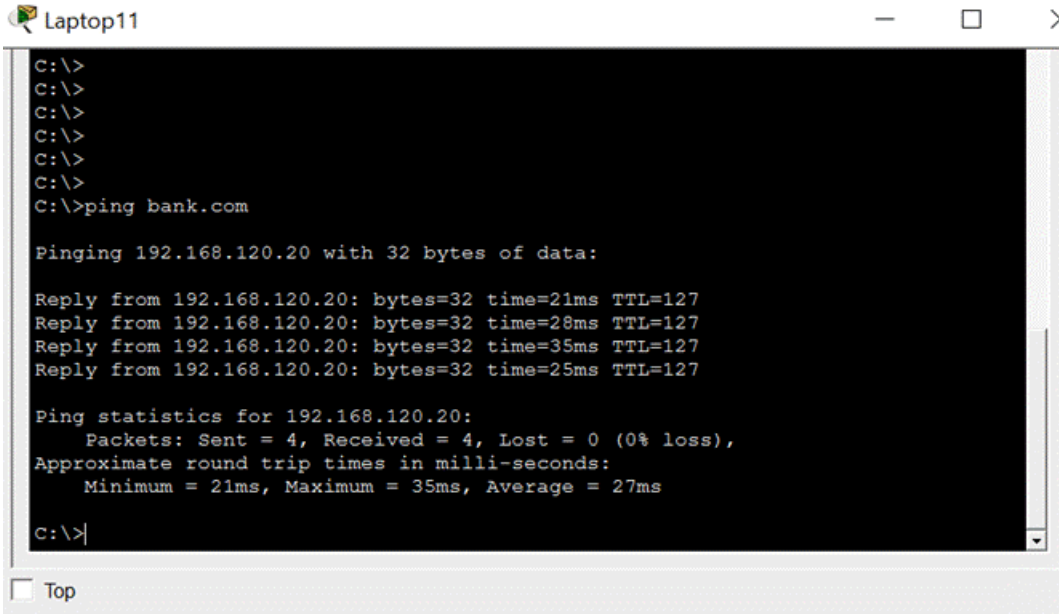

Figure 6.2 – DNS resolution test: bank.com resolved successfully from client VLAN

6.3 Web Server Access Test

Web server access was tested by launching the web browser from client devices in multiple VLANs and navigating to http://bank.com. The test confirms that inter-VLAN routing and NAT-forwarding are working, allowing the web page to load from any department.

Results: The bank's internal website loaded successfully from all tested VLANs, confirming correct routing from client VLANs through the multilayer switches and routers to the server in VLAN 120.

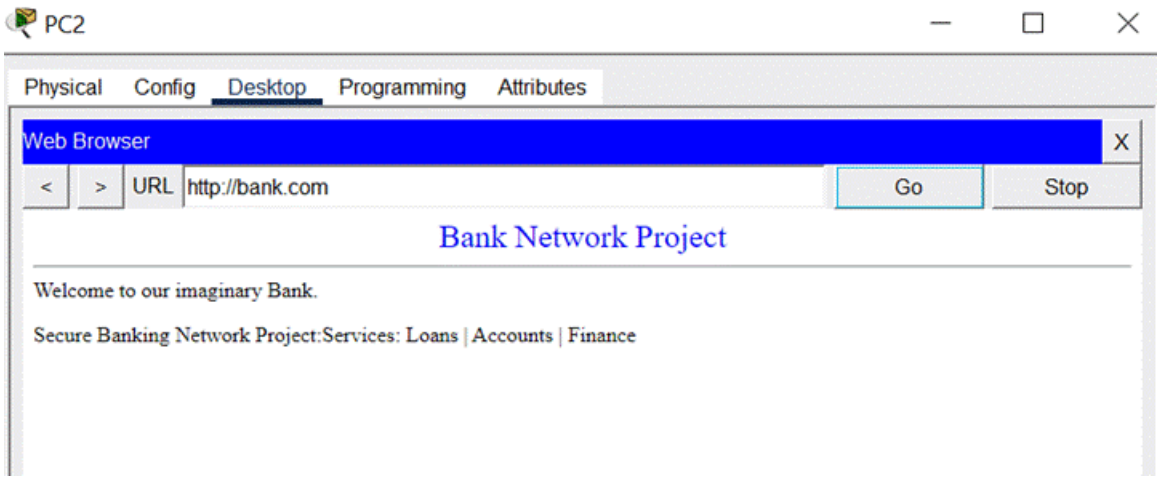

Figure 6.3 – Web server access test: http://bank.com loads from client in VLAN 40

6.4 FTP Server Test

FTP connectivity was tested using the Packet Tracer desktop FTP client on several end devices. A test login was performed to the FTP server at 192.168.120.30, and a sample file was transferred to verify the full functionality of the FTP service.

Results: FTP connection and file transfer succeeded from multiple VLANs, confirming that the FTP server is reachable and the inter-VLAN routing infrastructure correctly delivers traffic to VLAN 120.

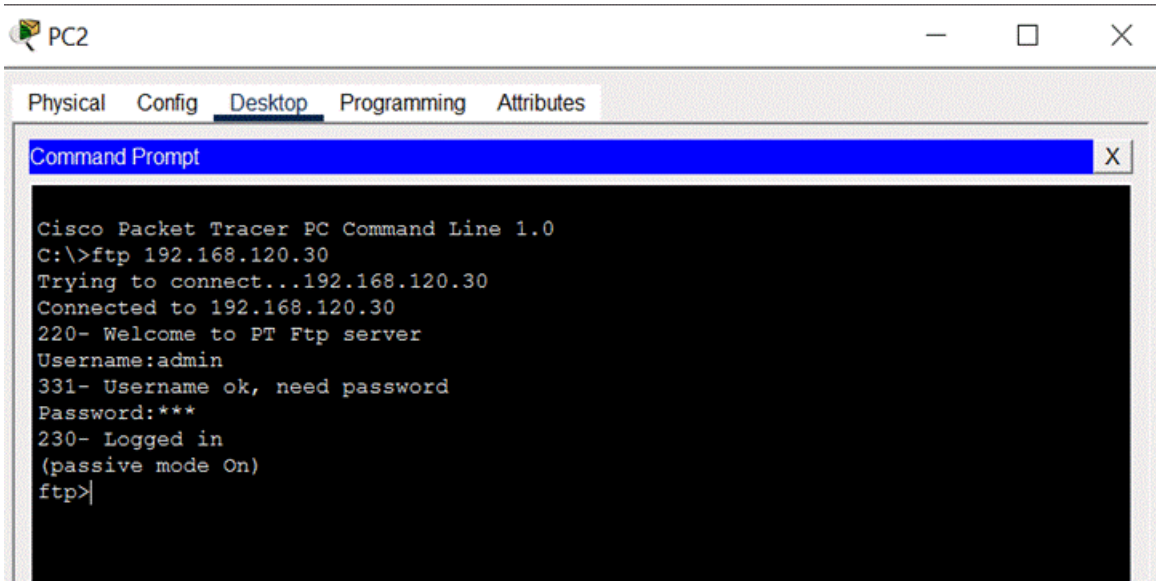

Figure 6.4 – FTP test: successful file transfer from client to FTP server at 192.168.120.30

6.5 Inter-VLAN Connectivity Test

Inter-VLAN routing was verified using ICMP ping tests between devices in different VLANs. These tests confirm that the SVIs (Switched Virtual Interfaces) on the multilayer switches are correctly configured and that routing between departments is working as intended.

Test cases performed:

- VLAN 10 (Management) → VLAN 50 (Accounting): Ping successful
- VLAN 30 (HR) → VLAN 60 (Finance): Ping successful
- VLAN 40 (Marketing) → VLAN 120 (Server Room): Ping successful
- VLAN 90 (Guest) → VLAN 120 (Server Room): Ping successful

The screenshots below show successful ping results and **VLAN-to-VLAN communication:**

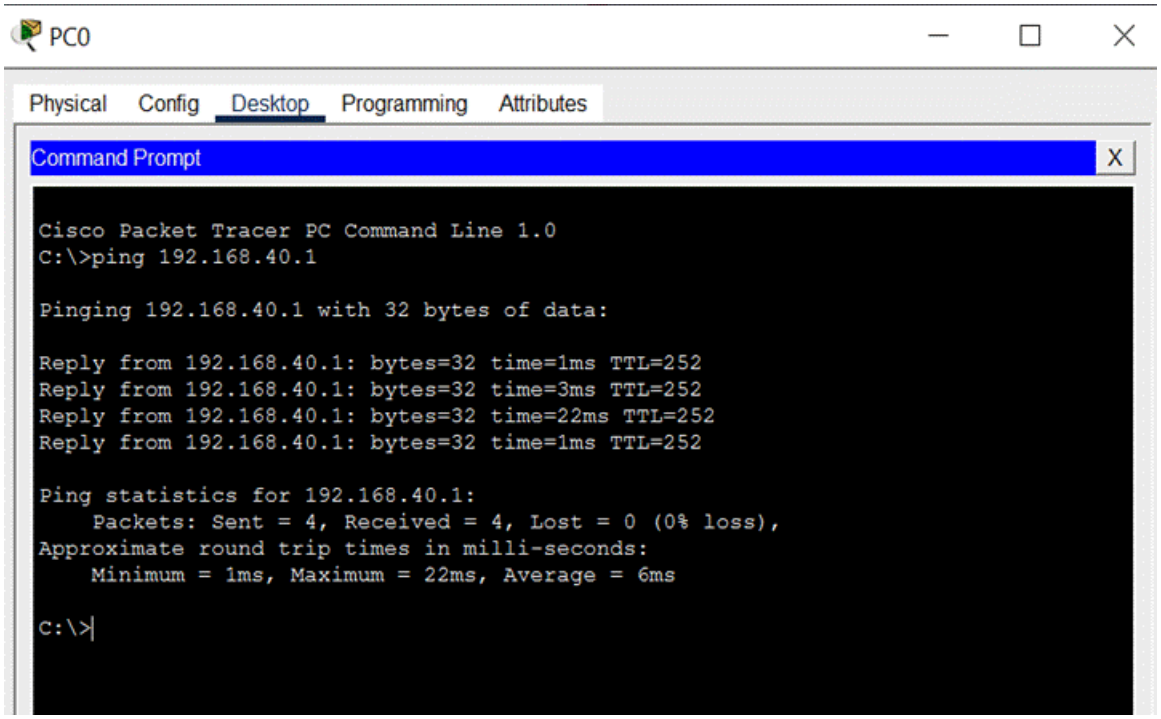

Figure 6.5a – Inter-VLAN ping test: VLAN 10 to VLAN 50 successful

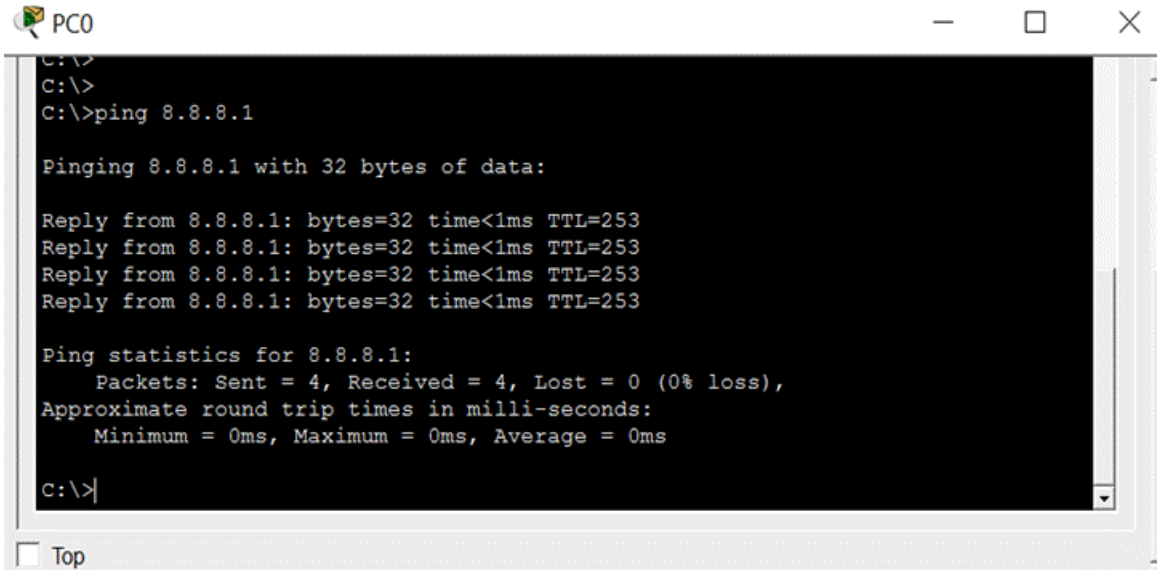

Figure 6.5b – Inter-VLAN ping test: cross-floor VLAN communication verified

6.6 Internet Access and NAT Test

Internet access through NAT was verified by pinging the external IP address (8.8.8.2) from internal devices in various VLANs. This test confirms that R1's NAT overload configuration correctly translates internal private addresses to the public IP (200.0.0.2) before forwarding traffic to the ISP.

Results: All internal VLANs successfully reached the external IP address (8.8.8.2), confirming that NAT is working correctly on R1 and that the default route to the ISP is properly propagated by OSPF to all routers and multilayer switches.

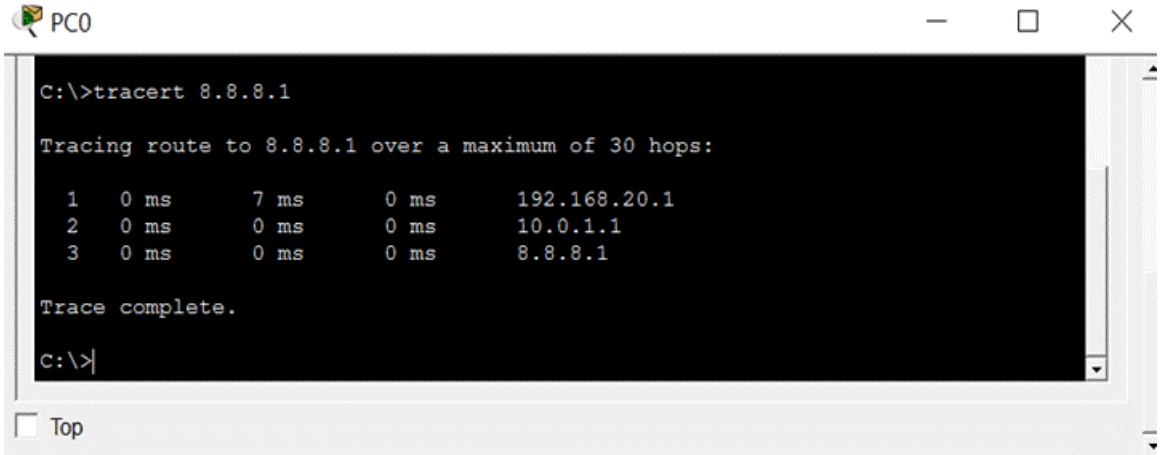

Figure 6.6 – Internet access test: ping to 8.8.8.2 from internal device succeeds via NAT

6.7 Route testing

The *'show ip route'* command is used to verify that the router can correctly identify and reach different networks. By examining the routing table, we confirm that all connected and remote network routes are present, ensuring that data can be properly forwarded across the network.

Results : All connected and remote network routes of the 3rd floor connected through a multilayer switch are present in the routing table along with OSPF routes that are successfully learned. This confirms that routing is correctly configured and operational.

```
Floor3-L3>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.3.1 to network 0.0.0.0

     10.0.0.0/30 is subnetted, 10 subnets
O       10.0.1.0 [110/66] via 10.0.3.1, 00:05:37, GigabitEthernet0/1
O       10.0.2.0 [110/66] via 10.0.3.1, 00:05:37, GigabitEthernet0/1
C       10.0.3.0 is directly connected, GigabitEthernet0/1
O       10.0.4.0 [110/66] via 10.0.3.1, 00:05:37, GigabitEthernet0/1
O       10.10.10.0 [110/129] via 10.0.3.1, 00:05:37, GigabitEthernet0/1
O       10.10.10.4 [110/65] via 10.0.3.1, 00:05:37, GigabitEthernet0/1
O       10.10.10.8 [110/129] via 10.0.3.1, 00:05:37, GigabitEthernet0/1
O       10.10.10.12 [110/129] via 10.0.3.1, 00:05:37, GigabitEthernet0/1
O       10.10.10.16 [110/65] via 10.0.3.1, 00:05:37, GigabitEthernet0/1
O       10.10.10.20 [110/65] via 10.0.3.1, 00:05:37, GigabitEthernet0/1
O    192.168.10.0/24 [110/67] via 10.0.3.1, 00:05:37, GigabitEthernet0/1
```

Figure 6.7a – 3rd Floor routing links through multilayer switch verified

The *'show ip interface brief'* command is used to verify that all router interfaces are properly configured and operational. By checking the interface status (up/up) and assigned IP addresses, we confirm that the physical and logical connections of the router are active and ready for communication.

Results : All required interfaces are up/up. IP addresses are correctly configured on each interface.This confirms that the router interfaces are active and functioning properly.

```
Floor3-L3>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES unset  up                    up
FastEthernet0/2        unassigned      YES unset  up                    up
FastEthernet0/3        unassigned      YES unset  up                    up
FastEthernet0/4        unassigned      YES unset  down                  down
FastEthernet0/5        unassigned      YES unset  down                  down
FastEthernet0/6        unassigned      YES unset  down                  down
FastEthernet0/7        unassigned      YES unset  down                  down
FastEthernet0/8        unassigned      YES unset  down                  down
FastEthernet0/9        unassigned      YES unset  down                  down
FastEthernet0/10       unassigned      YES unset  down                  down
FastEthernet0/11       unassigned      YES unset  down                  down
FastEthernet0/12       unassigned      YES unset  down                  down
FastEthernet0/13       unassigned      YES unset  down                  down
FastEthernet0/14       unassigned      YES unset  down                  down
FastEthernet0/15       unassigned      YES unset  down                  down
FastEthernet0/16       unassigned      YES unset  down                  down
FastEthernet0/17       unassigned      YES unset  down                  down
FastEthernet0/18       unassigned      YES unset  down                  down
FastEthernet0/19       unassigned      YES unset  down                  down
FastEthernet0/20       unassigned      YES unset  down                  down
FastEthernet0/21       unassigned      YES unset  down                  down
FastEthernet0/22       unassigned      YES unset  down                  down
FastEthernet0/23       unassigned      YES unset  down                  down
FastEthernet0/24       unassigned      YES unset  down                  down
GigabitEthernet0/1     10.0.3.2        YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan70                 192.168.70.1    YES manual up                    up
Vlan80                 192.168.80.1    YES manual up                    up
Vlan90                 192.168.90.1    YES manual up                    up
Floor3-L3>
Floor3-L3>
```

Figure 6.7b –Showing all router interfaces in up/up state with correct IP configuration

6.8 Test Summary

NO	Test	Description	Expected	Result
1	DHCP Assignment	Devices auto-receive IP in correct VLAN subnet	IP assigned	Pass
2	DNS Resolution	bank.com resolves to 192.168.120.20	Name resolved	Pass
3	Web Server Access	http://bank.com loads from any VLAN	Page loads	Pass
4	FTP File Transfer	Login + transfer to 192.168.120.30	File transfer OK	Pass

5	Inter-VLAN Ping	Ping between different department VLANs	Ping success	Pass
6	NAT / Internet	Ping 8.8.8.2 from internal device	Ping success	Pass
7	Simulation Mode	Trace packet path end-to-end	Correct path	Pass
8	Route Testing	Interfaces up/up with correct IPs and all network routes present.	Correct connections	Pass

7. Conclusion

7.1 Summary of the Network Design

This project successfully designed and simulated a full-scale bank enterprise network using Cisco Packet Tracer. The network was built to meet the operational, security, and scalability requirements of a modern banking institution. Spanning four floors with twelve departments, the design integrates multiple networking technologies into a unified, hierarchical, and manageable infrastructure.

Key achievements of the design include:

- **Hierarchical Architecture:** The network follows a three-tier hierarchy consisting of the core layer (routers and inter-floor WAN links), distribution layer (Layer 3 multilayer switches per floor), and access layer (Layer 2 switches per department). This structure ensures scalability, easier management, and high performance.
- **VLAN Segmentation:** Twelve VLANs (VLAN 10 through VLAN 120) were created, one per department, to logically isolate traffic, reduce broadcast domains, improve security, and enable department-level access control.
- **IP Addressing:** A structured IP addressing scheme was applied using /24 subnets (192.168.X.0/24) for departmental VLANs, /30 subnets (10.10.10.x/30 and 10.0.x.0/30) for point-to-point router and switch links, and a public /30 subnet (200.0.0.0/30) for the ISP connection.

- **Dynamic Routing with OSPF:** OSPF was deployed across all routers and multilayer switches to enable automatic route propagation, fast convergence, and optimal path selection between floors and departments.
- **Centralized Network Services:** Three servers in the Server Room (VLAN 120) provide DHCP, DNS, web (bank.com), and FTP services to all departments, ensuring reliable and centralized delivery of essential banking applications.
- **NAT for Internet Access:** PAT (NAT Overload) on Router R1 enables all internal devices to access the internet through a single public IP address, conserving public address space and adding a layer of security.
- **Wireless Coverage:** Access points are deployed in each department to support mobile staff using smartphones and tablets, ensuring mobility and flexibility throughout the building.

All components were successfully configured and verified through comprehensive testing. DHCP assignment, DNS resolution, web and FTP access, inter-VLAN communication, and internet connectivity were all confirmed to be functioning correctly in Cisco Packet Tracer.

7.2 Challenges Faced and How They Were Solved

Challenge 1: DHCP Requests Not Reaching the Multilayer Switch

Initially, devices in certain VLANs were not receiving DHCP-assigned IP addresses. This was because DHCP uses broadcast messages that do not cross Layer 3 boundaries by default.

Solution: DHCP pools were configured directly on each floor's multilayer switch, with a separate pool per VLAN using the network command. Since the SVIs on the same switch serve as default gateways for those VLANs, DHCP broadcasts from devices are handled locally on the switch without needing relay agents. This resolved all DHCP failures.

Challenge 2: Inter-VLAN Communication Failures

After configuring VLANs on access switches, devices in different VLANs were unable to communicate. Pings between departments were failing even though routing was enabled on the multilayer switches.

Solution: The root cause was missing or incorrectly configured SVIs (Switched Virtual Interfaces) on the multilayer switches. Each VLAN required a corresponding SVI with the correct IP address to act as the default gateway. After properly configuring all SVIs and verifying the ip routing command was active, inter-VLAN routing functioned correctly.

Challenge 3: VLAN Traffic Not Passing Between Switches

VLAN traffic was being dropped between access switches and the multilayer switches because the inter-switch links were in access mode rather than trunk mode.

Solution: All uplinks from access switches to the multilayer switches were reconfigured as 802.1Q trunk ports using switchport mode trunk and, where required for the Cisco 3560, the switchport trunk encapsulation dot1q command. This allowed multiple VLAN traffic to pass over a single physical uplink.

Challenge 4: NAT Not Translating Internal Traffic

After configuring NAT on R1, internal devices were still unable to reach the external IP (8.8.8.2). The NAT translations table showed no active entries.

Solution: Investigation revealed that the inside and outside NAT designations were missing from the interfaces. The ip nat inside command was added to G0/0 and ip nat outside to G0/1. Additionally, the ACL used by NAT was too restrictive and only matched one subnet. It was updated with a wildcard mask of 0.0.255.255 to include the entire 192.168.0.0/16 address space. After these corrections, NAT translations appeared correctly and internet access was confirmed.

7.3 Final Remarks

The completion of this project demonstrates a practical and comprehensive application of enterprise networking concepts in a real-world banking context. From initial design through implementation and testing, every phase of the project reinforced core principles including IP addressing and subnetting, VLAN segmentation, dynamic routing, and network services deployment.

The bank enterprise network as designed in Cisco Packet Tracer is production-ready in terms of its logical architecture. It provides secure department isolation through VLANs, reliable communication via OSPF, centralized services through the server room infrastructure, and internet access through NAT. The hierarchical design ensures the network can be scaled in the future by adding new floors, departments, or server capacity without significant redesign.

Working as part of a team on this project also highlighted the importance of consistent documentation, coordinated IP address planning, and systematic testing. Each member's contribution was integral to the overall success of the design, and the division of tasks across sections helped manage the complexity of a large-scale enterprise topology.